

30.1 Penalized least squares in general

Ridge regression and the lasso of [Chapter 27](#) raise the same questions: is there a minimizer, is it unique, and how do we recognize one? This section answers them for a general, mainly convex, penalty, reads a penalty as a prior, and ends with penalties that give up convexity on purpose. Throughout, $\mathbf{X}$ is $n \times p$ of any rank, and p may exceed n .

30.1.1 The criterion

Definition 30.1.1 — Penalized least squares

A **penalty** is a function $P : \mathbb{R}^p \rightarrow [0, \infty)$. For $\lambda \geq 0$, the **penalized least squares criterion** is

$$Q_\lambda(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda P(\mathbf{b}), \quad (30.1.1)$$

and a **penalized least squares estimate** is any minimizer $\hat{\beta}_\lambda$ of Q_λ . The penalty is **convex** if $P(\alpha\mathbf{b} + (1 - \alpha)\mathbf{c}) \leq \alpha P(\mathbf{b}) + (1 - \alpha)P(\mathbf{c})$ for all $\mathbf{b}, \mathbf{c}$ and $\alpha \in (0, 1)$, and **strictly convex** if the inequality is strict whenever $\mathbf{b} \neq \mathbf{c}$. The penalization is **partial** if $\mathbf{b} = (\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$ and P depends on $\mathbf{b}_2$ alone; the coefficients in $\mathbf{b}_1$ are then **unpenalized**.

Other scalings (no $\frac{1}{2}$, or a divisor n) multiply λ by a constant; results are quoted here in the scaling of [Eq. 30.1.1](#). Ridge ($P = \frac{1}{2} \|\mathbf{b}\|^2$, [Theorem 27.2.2](#)) and the lasso ($P = \|\mathbf{b}\|_1$, [Definition 27.4.1](#)) are convex; the best-subset penalty $\#\{j : b_j \neq 0\}$ of [Chapter 29](#) and the bridge penalties $\sum_j |b_j|^q$, $q < 1$, are not. [Sections 30.2 and 30.3](#) add penalties with structure.

Intercept and scaling. An intercept is not penalized. Minimizing over it first gives $b_0 = \bar{y} - \bar{\mathbf{x}}^\top \mathbf{b}$ and leaves [Eq. 30.1.1](#) with $\mathbf{y}$ and the columns of $\mathbf{X}$ centred ([Exercise \(B2\)](#)), the partialling-out of [Theorem 6.6.2](#), so we centre and drop it. A penalty that treats coordinates alike is not equivariant under rescaling a column, so the columns are also scaled to a common length, a choice that is part of the model.

30.1.2 Existence and uniqueness

For least squares a minimizer always exists and its fit is unique ([Theorem 6.4.1](#)). For penalized criteria:

Theorem 30.1.1 — Existence and uniqueness of penalized estimates

Let P be continuous and $\lambda \geq 0$.

- (a) If $Q_\lambda(\mathbf{b}) \rightarrow \infty$ as $\|\mathbf{b}\| \rightarrow \infty$, a minimizer exists. This holds in each of three cases:
 - (i) $\mathbf{X}$ has full column rank; (ii) $\lambda > 0$ and $P(\mathbf{b}) \geq c\|\mathbf{b}\|$ for some $c > 0$; (iii) the penalization is partial, $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ with $\mathbf{X}_1$ of full column rank, $\lambda > 0$, and

$P(\mathbf{b}) \geq c\|\mathbf{b}_2\|$ for some $c > 0$.

- (b) If P is convex, the set of minimizers is convex, and all minimizers have the same fitted vector $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda$. If in addition $\lambda > 0$, they all have the same penalty $P(\hat{\boldsymbol{\beta}}_\lambda)$.
- (c) If P is convex and either $\mathbf{X}$ has full column rank, or $\lambda > 0$ and P is strictly convex, the minimizer is unique.

Proof.

- (a) Let $\mathcal{S} = \{\mathbf{b} : Q_\lambda(\mathbf{b}) \leq Q_\lambda(\mathbf{0})\}$. It is closed because Q_λ is continuous, and bounded because $Q_\lambda \rightarrow \infty$, so Q_λ attains its minimum on $\mathcal{S}$. Any point outside $\mathcal{S}$ has a larger value than $\mathbf{0}$, so this minimum is a global minimizer. In case (i), the smallest singular value s of $\mathbf{X}$ is positive (Theorem 1.8.1), so $\|\mathbf{y} - \mathbf{X}\mathbf{b}\| \geq s\|\mathbf{b}\| - \|\mathbf{y}\|$, which tends to infinity. In case (ii), $Q_\lambda(\mathbf{b}) \geq \lambda c\|\mathbf{b}\|$. In case (iii), suppose $\|\mathbf{b}^{(k)}\| \rightarrow \infty$ along a sequence on which Q_λ stays bounded. Then $\lambda c\|\mathbf{b}_2^{(k)}\|$ is bounded, so $\|\mathbf{b}_1^{(k)}\| \rightarrow \infty$. With $s_1 > 0$ the smallest singular value of $\mathbf{X}_1$, $\|\mathbf{y} - \mathbf{X}\mathbf{b}^{(k)}\| \geq s_1\|\mathbf{b}_1^{(k)}\| - \|\mathbf{y}\| - \|\mathbf{X}_2\mathbf{b}_2^{(k)}\| \rightarrow \infty$, a contradiction.
- (b) If P is convex, so is Q_λ , as a sum of convex functions, and the set of minimizers of a convex function is convex. Let $\mathbf{b}$ and $\mathbf{c}$ be minimizers with value q , and put $\mathbf{m} = \frac{1}{2}(\mathbf{b} + \mathbf{c})$. With $\mathbf{u} = \mathbf{y} - \mathbf{X}\mathbf{b}$ and $\mathbf{v} = \mathbf{y} - \mathbf{X}\mathbf{c}$, the parallelogram identity gives

$$\|\mathbf{y} - \mathbf{X}\mathbf{m}\|^2 = \left\|\frac{1}{2}(\mathbf{u} + \mathbf{v})\right\|^2 = \frac{1}{2}\|\mathbf{u}\|^2 + \frac{1}{2}\|\mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{X}(\mathbf{b} - \mathbf{c})\|^2.$$

Together with $P(\mathbf{m}) \leq \frac{1}{2}P(\mathbf{b}) + \frac{1}{2}P(\mathbf{c})$ this yields $q \leq Q_\lambda(\mathbf{m}) \leq q - \frac{1}{8}\|\mathbf{X}(\mathbf{b} - \mathbf{c})\|^2$. Hence $\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{c}$. The two minimizers then have equal residual sums of squares, so when $\lambda > 0$ their penalties are equal too.

- (c) If $\mathbf{X}$ has full column rank, $\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{c}$ forces $\mathbf{b} = \mathbf{c}$. If P is strictly convex and $\mathbf{b} \neq \mathbf{c}$, the inequality for $P(\mathbf{m})$ in (b) is strict and $Q_\lambda(\mathbf{m}) < q$, which is impossible. ■

Norms satisfy (a)(ii), so the lasso, group lasso and elastic net always have minimizers. By (b), when $p > n$ the lasso's minimizers, possibly many, share their fit and ℓ_1 norm (they are unique in general position; Tibshirani, 2013).

30.1.3 Optimality conditions

The corner of $|b|$ at zero, which makes the lasso set coefficients to zero, calls for a substitute for the gradient.

Definition 30.1.2 — Subgradient

Let P be convex. A vector $\mathbf{s}$ is a **subgradient** of P at $\mathbf{b}$ if $P(\mathbf{c}) \geq P(\mathbf{b}) + \mathbf{s}^\top(\mathbf{c} - \mathbf{b})$ for every $\mathbf{c} \in \mathbb{R}^p$. The set of all subgradients at $\mathbf{b}$ is the **subdifferential** $\partial P(\mathbf{b})$.

Where P is differentiable, $\partial P(\mathbf{b}) = \{\nabla P(\mathbf{b})\}$. For $P(b) = |b|$ on $\mathbb{R}$, the subdifferential is $\{\text{sign}(b)\}$ if $b \neq 0$ and $[-1, 1]$ at 0.

Lemma 30.1.2 — Optimality condition

Let P be convex and $\lambda > 0$. A vector $\hat{\mathbf{b}}$ minimizes Q_λ iff

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}) \in \lambda \partial P(\hat{\mathbf{b}}). \quad (30.1.2)$$

In that case $Q_\lambda(\mathbf{b}) - Q_\lambda(\hat{\mathbf{b}}) \geq \frac{1}{2}\|\mathbf{X}(\mathbf{b} - \hat{\mathbf{b}})\|^2$ for every $\mathbf{b}$.

Proof. Write $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\mathbf{b}}$ and $\mathbf{d} = \mathbf{b} - \hat{\mathbf{b}}$. Expanding the square,

$$Q_\lambda(\mathbf{b}) - Q_\lambda(\hat{\mathbf{b}}) = -\mathbf{d}^\top \mathbf{X}^\top \mathbf{r} + \frac{1}{2}\|\mathbf{X}\mathbf{d}\|^2 + \lambda(P(\mathbf{b}) - P(\hat{\mathbf{b}})). \quad (30.1.3)$$

If $\mathbf{X}^\top \mathbf{r} = \lambda \mathbf{s}$ with $\mathbf{s} \in \partial P(\hat{\mathbf{b}})$, then $\lambda(P(\mathbf{b}) - P(\hat{\mathbf{b}})) \geq \lambda \mathbf{s}^\top \mathbf{d} = \mathbf{d}^\top \mathbf{X}^\top \mathbf{r}$, and [Eq. 30.1.3](#) gives the stated lower bound, which is nonnegative. Conversely, let $\hat{\mathbf{b}}$ be a minimizer and fix $\mathbf{b}$. For $0 < \tau < 1$, convexity gives $P(\hat{\mathbf{b}} + \tau \mathbf{d}) - P(\hat{\mathbf{b}}) \leq \tau(P(\mathbf{b}) - P(\hat{\mathbf{b}}))$. Apply [Eq. 30.1.3](#) to $\hat{\mathbf{b}} + \tau \mathbf{d}$ in place of $\mathbf{b}$:

$$\begin{aligned} 0 &\leq Q_\lambda(\hat{\mathbf{b}} + \tau \mathbf{d}) - Q_\lambda(\hat{\mathbf{b}}) \\ &\leq -\tau \mathbf{d}^\top \mathbf{X}^\top \mathbf{r} + \frac{\tau^2}{2}\|\mathbf{X}\mathbf{d}\|^2 + \lambda \tau(P(\mathbf{b}) - P(\hat{\mathbf{b}})). \end{aligned}$$

Divide by τ and let $\tau \rightarrow 0$. The result is $\lambda(P(\mathbf{b}) - P(\hat{\mathbf{b}})) \geq \mathbf{d}^\top \mathbf{X}^\top \mathbf{r}$ for every $\mathbf{b}$, which says that $\mathbf{X}^\top \mathbf{r} / \lambda \in \partial P(\hat{\mathbf{b}})$. ■

Condition [Eq. 30.1.2](#) is the **Karush–Kuhn–Tucker** (KKT) condition; with $\lambda = 0$ it is the normal equations. Most penalties below are sums of Euclidean norms of blocks.

Lemma 30.1.3 — Subdifferential of a sum of block norms

Let $G_1, \dots, G_K$ partition $\{1, \dots, p\}$, let $w_1, \dots, w_K > 0$, and let $P(\mathbf{b}) = \sum_k w_k \|\mathbf{b}_{G_k}\|$, where $\mathbf{b}_{G_k}$ is the subvector of $\mathbf{b}$ with coordinates in G_k . Then $\mathbf{s} \in \partial P(\mathbf{b})$ iff, for every k ,

$$\mathbf{s}_{G_k} = w_k \frac{\mathbf{b}_{G_k}}{\|\mathbf{b}_{G_k}\|} \quad \text{if } \mathbf{b}_{G_k} \neq \mathbf{0}, \quad \|\mathbf{s}_{G_k}\| \leq w_k \quad \text{if } \mathbf{b}_{G_k} = \mathbf{0}.$$

With singleton blocks and unit weights, $P = \|\cdot\|_1$, and $\mathbf{s} \in \partial \|\mathbf{b}\|_1$ iff $s_j = \text{sign}(b_j)$ when $b_j \neq 0$ and $|s_j| \leq 1$ when $b_j = 0$.

Proof. First, $\mathbf{s} \in \partial P(\mathbf{b})$ iff $\mathbf{s}_{G_k} \in \partial(w_k \|\cdot\|)(\mathbf{b}_{G_k})$ for every k . Summing the blockwise inequalities gives one direction. For the other, apply the subgradient inequality to a $\mathbf{c}$ that differs from $\mathbf{b}$ only in block k . So it suffices to describe the subdifferential of $f(\mathbf{u}) = w\|\mathbf{u}\|$ on $\mathbb{R}^d$. We show that $\mathbf{s} \in \partial f(\mathbf{u})$ iff $\|\mathbf{s}\| \leq w$ and $\mathbf{s}^\top \mathbf{u} = w\|\mathbf{u}\|$.

If these two conditions hold, then for every $\mathbf{v}$, by Cauchy–Schwarz ([Proposition 1.1.1](#)), $f(\mathbf{v}) = w\|\mathbf{v}\| \geq \mathbf{s}^\top \mathbf{v} = f(\mathbf{u}) + \mathbf{s}^\top (\mathbf{v} - \mathbf{u})$. Conversely, let $\mathbf{s} \in \partial f(\mathbf{u})$. Taking $\mathbf{v} = \mathbf{0}$ and $\mathbf{v} = 2\mathbf{u}$ gives $\mathbf{s}^\top \mathbf{u} \geq w\|\mathbf{u}\|$ and $\mathbf{s}^\top \mathbf{u} \leq w\|\mathbf{u}\|$. Then $w\|\mathbf{u} + \tau \mathbf{v}\| \geq w\|\mathbf{u}\| + \tau \mathbf{s}^\top \mathbf{v}$ for all $\tau > 0$. The triangle inequality bounds the left side by $w\|\mathbf{u}\| + \tau w\|\mathbf{v}\|$, so $\mathbf{s}^\top \mathbf{v} \leq w\|\mathbf{v}\|$

for every $\mathbf{v}$. Taking $\mathbf{v} = \mathbf{s}$ gives $\|\mathbf{s}\| \leq w$.

If $\mathbf{u} = \mathbf{0}$, the two conditions say $\|\mathbf{s}\| \leq w$. If $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{s}^\top \mathbf{u} = w\|\mathbf{u}\| \geq \|\mathbf{s}\|\|\mathbf{u}\|$ is equality in Cauchy–Schwarz, so $\mathbf{s}$ is a nonnegative multiple of $\mathbf{u}$ with norm w , that is, $\mathbf{s} = w\mathbf{u}/\|\mathbf{u}\|$. ■

Combining the two lemmas, $\hat{\boldsymbol{\beta}}$ is a lasso estimate iff $\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \lambda \text{sign}(\hat{\beta}_j)$ for every j with $\hat{\beta}_j \neq 0$ and $|\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})| \leq \lambda$ for every j with $\hat{\beta}_j = 0$, as in Definition 27.4.1: active regressors share the absolute correlation λ with the residual, and no inactive one exceeds it.

30.1.4 The Bayesian reading

Ridge regression is a posterior mean (Proposition 7.6.3(d)). For other penalties the matching statement is about the posterior *mode*; part (b) below restates, for completeness, the reading of the lasso found in Exercise (B2) (Section 27.4).

Proposition 30.1.4 — Penalized estimates as posterior modes

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ with σ^2 known, and give $\boldsymbol{\beta}$ a prior density $\pi(\mathbf{b}) \propto \exp\{-(\lambda/\sigma^2)P(\mathbf{b})\}$, assumed integrable. Then the posterior density of $\boldsymbol{\beta}$ is proportional to $\exp\{-Q_\lambda(\mathbf{b})/\sigma^2\}$. So the penalized least squares estimates are exactly the posterior modes. In particular:

- (a) a normal prior $\boldsymbol{\beta} \sim N_p(\mathbf{0}, \tau^2 \mathbf{I})$ gives ridge regression (Theorem 27.2.2) with $\lambda = \sigma^2/\tau^2$, and the posterior mode equals the posterior mean;
- (b) independent Laplace priors with density $(a/2)e^{-a|b_j|}$ give the lasso with $\lambda = a\sigma^2$.

Proof. By Bayes' theorem the posterior density is proportional to the likelihood times the prior, that is, to $\exp\{-\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2/(2\sigma^2)\} \exp\{-(\lambda/\sigma^2)P(\mathbf{b})\} = \exp\{-Q_\lambda(\mathbf{b})/\sigma^2\}$, and $t \mapsto e^{-t/\sigma^2}$ is decreasing. For (a), $\tau^{-2}\|\mathbf{b}\|^2/2 = (\lambda/\sigma^2)\|\mathbf{b}\|^2/2$ with $\lambda = \sigma^2/\tau^2$. The posterior is normal, so its mode and mean coincide (Theorem 7.6.2 with σ^2 known, or Proposition 7.6.3(d)). For (b), as in Exercise (B2), $a\|\mathbf{b}\|_1 = (\lambda/\sigma^2)\|\mathbf{b}\|_1$ with $\lambda = a\sigma^2$. ■

Under squared error loss, though, a Bayesian reports the posterior mean, which under a Laplace prior is never sparse.

Example. (Posterior mode and posterior mean under a Laplace prior)

Take one observation $z \sim N(\theta, 1)$ and a Laplace prior on θ with rate $a = 1.5$. By Proposition 30.1.4(b) the posterior mode is the lasso estimate with $\lambda = 1.5$, which is soft thresholding at 1.5 (Theorem 27.4.2). The posterior mean, computed by numerical integration in the listing below, behaves differently:

z	0.5	1.0	2.0	4.0
posterior mode	0	0	0.500	2.500

posterior mean 0.174 0.364 0.859 2.509

The mode is zero for $|z| \leq 1.5$; the mean is zero only at $z = 0$, and approaches $z - 1.5$ for large z (Figure 30.1.1(c)). Exact zeros come from the mode, not from the prior; a prior that believes in exact zeros needs a point mass, as in the spike-and-slab priors of George and McCulloch (1993). Chapter 39 returns to priors as regularizers (Proposition 39.5.1).

```
import numpy as np
from scipy import integrate

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)

def laplace_posterior_mean(z, rate):
    """E(theta | z) when z ~ N(theta, 1) and theta has density (rate/2) exp(-rate / theta)."""
    w = lambda t: np.exp(-0.5 * (z - t) ** 2 - rate * abs(t))
    num = integrate.quad(lambda t: t * w(t), -30, 30, points=[0.0, z])[0]
    den = integrate.quad(w, -30, 30, points=[0.0, z])[0]
    return num / den

rate = 1.5
for z in [0.5, 1.0, 2.0, 4.0]:
    print(f"z = {z}: posterior mode {float(soft(z, rate)):.3f}, "
          f"posterior mean {laplace_posterior_mean(z, rate):.3f}")
```

[Run this code online](#)

30.1.5 Penalties that are not convex

Under an orthonormal design the lasso subtracts λ from every surviving coefficient, however large (Theorem 27.4.2). Fan and Li (2001) asked for a rule that is **sparse**, **continuous** in the data, and **nearly unbiased** (large estimates left alone). Soft thresholding fails the third, and hard thresholding, the best-subset rule (Theorem 27.4.2(c)), the second. No convex penalty has all three, since one that is flat far from zero on both sides is constant. Two penalties $P(\mathbf{b}) = \sum_j \rho(b_j)$, with ρ concave on $[0, \infty)$, are in common use.

SCAD (the smoothly clipped absolute deviation of Fan and Li, 2001), with a parameter $a > 2$, is defined through its derivative on $t > 0$:

$$\rho'(t) = \lambda \left\{ \mathbf{1}(t \leq \lambda) + \frac{(a\lambda - t)_+}{(a-1)\lambda} \mathbf{1}(t > \lambda) \right\}, \quad \rho(0) = 0, \quad \rho(-t) = \rho(t).$$

It is the lasso penalty up to λ ; its slope then falls linearly to zero at $a\lambda$. Here λ sits inside ρ , and the criterion is $\frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \sum_j \rho(b_j)$. Fan and Li suggest $a = 3.7$.

MCP (the minimax concave penalty of Zhang, 2010), with a parameter $\gamma > 1$, is

$$\rho(t) = \begin{cases} \lambda|t| - t^2/(2\gamma), & |t| \leq \gamma\lambda, \\ \gamma\lambda^2/2, & |t| > \gamma\lambda. \end{cases}$$

Its slope $\lambda - |t|/\gamma$ starts to fall at once, rather than after λ as for SCAD.

Proposition 30.1.5 — MCP: convexity and firm thresholding

Let $Q(\mathbf{b}) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \sum_j \rho(b_j)$ with ρ the MCP penalty.

- (a) If the smallest eigenvalue of $\mathbf{X}^\top \mathbf{X}$ is at least $1/\gamma$, then Q is convex. If it exceeds $1/\gamma$, then Q is strictly convex and has a unique minimizer.
- (b) If $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, the unique minimizer is $\hat{\beta}_j = \phi(z_j)$ with $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$, where ϕ is **firm thresholding**:

$$\phi(z) = \begin{cases} 0, & |z| \leq \lambda, \\ \text{sign}(z) \frac{|z| - \lambda}{1 - 1/\gamma}, & \lambda < |z| \leq \gamma\lambda, \\ z, & |z| > \gamma\lambda. \end{cases}$$

Proof.

- (a) Write

$$Q(\mathbf{b}) = \left\{ \frac{1}{2} \mathbf{b}^\top (\mathbf{X}^\top \mathbf{X} - \gamma^{-1} \mathbf{I}) \mathbf{b} - \mathbf{y}^\top \mathbf{X} \mathbf{b} + \frac{1}{2} \mathbf{y}^\top \mathbf{y} \right\} + \sum_j \left\{ \rho(b_j) + \frac{b_j^2}{2\gamma} \right\}.$$

The first brace is a quadratic with nonnegative definite matrix, hence convex, and strictly convex if the matrix is positive definite. Each term of the second sum is $\lambda|t|$ for $|t| \leq \gamma\lambda$ and $\gamma\lambda^2/2 + t^2/(2\gamma)$ beyond. On $t > 0$ its derivative is λ up to $\gamma\lambda$ and t/γ after. This derivative is continuous at $\gamma\lambda$ and nondecreasing, so the function is convex (it is even, with a corner at 0). A sum of convex functions is convex, and adding a strictly convex function makes the sum strictly convex. A strictly convex function that tends to infinity has exactly one minimizer, by the argument of [Theorem 30.1.1](#)(a) and (c).

- (b) With $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, part (a) applies because $\gamma > 1$. Moreover $Q(\mathbf{b}) = \frac{1}{2}\|\mathbf{z} - \mathbf{b}\|^2 + \sum_j \rho(b_j) + \text{const}$, which separates into one problem per coordinate: minimize $g(\theta) = \frac{1}{2}(z - \theta)^2 + \rho(\theta)$. Take $z \geq 0$. For $\theta > 0$, $g(-\theta) - g(\theta) = 2z\theta \geq 0$, so we may restrict to $\theta \geq 0$. There

$$g'(\theta) = \begin{cases} (1 - 1/\gamma)\theta - (z - \lambda), & 0 < \theta \leq \gamma\lambda, \\ \theta - z, & \theta > \gamma\lambda, \end{cases}$$

a continuous nondecreasing function. If $z \leq \lambda$, then $g' \geq 0$ on $(0, \infty)$, and the minimizer is 0. If $\lambda < z \leq \gamma\lambda$, then g' vanishes at $\theta = (z - \lambda)/(1 - 1/\gamma)$, which is at most $\gamma\lambda$. If $z > \gamma\lambda$, then $g'(\gamma\lambda) = \gamma\lambda - z < 0$, and g' vanishes at $\theta = z$.

■

Firm thresholding has all three properties; it tends to soft thresholding as $\gamma \rightarrow \infty$ and to hard thresholding as $\gamma \downarrow 1$. The SCAD rule under an orthonormal design is (Fan and Li, 2001)

$$\phi_{\text{SCAD}}(z) = \begin{cases} \text{sign}(z)(|z| - \lambda)_+, & |z| \leq 2\lambda, \\ \frac{(a-1)z - \text{sign}(z)a\lambda}{a-2}, & 2\lambda < |z| \leq a\lambda, \\ z, & |z| > a\lambda, \end{cases}$$

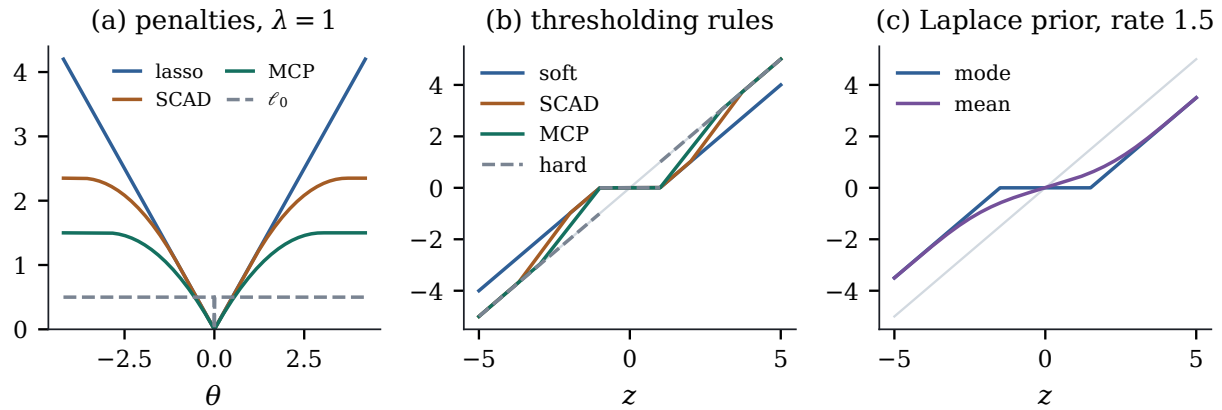


Figure 30.1.1: (a) The lasso, SCAD ($a = 3.7$), MCP ($\gamma = 3$) and ℓ_0 penalties with $\lambda = 1$. (b) The thresholding rules they induce under an orthonormal design (soft, SCAD, firm, and hard thresholding dashed); the grey line is the identity. (c) Posterior mode and posterior mean of θ given $z \sim N(\theta, 1)$ under a Laplace prior with rate 1.5 ([Example \(Posterior mode and posterior mean under a Laplace prior\)](#)).

which [Exercise \(B1\)](#) derives. The script checks each rule of [Figure 30.1.1](#) against brute-force minimization.

```
import numpy as np

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)

def mcp_rule(z, lam, gam):
    # firm thresholding
    az = np.abs(z)
    mid = np.sign(z) * (az - lam) / (1 - 1 / gam)
    return np.where(az <= lam, 0.0, np.where(az <= gam * lam, mid, z))

def scad_rule(z, lam, a):
    az = np.abs(z)
    mid = ((a - 1) * z - np.sign(z) * a * lam) / (a - 2)
    return np.where(az <= 2 * lam, soft(z, lam), np.where(az <= a * lam, mid, z))

def hard_rule(z, lam):
    # penalty (lam^2 / 2) * 1{theta != 0}
    return np.where(np.abs(z) > lam, z, 0.0)
```

[Run this code online](#)

Without the eigenvalue condition there can be several local minima ([Exercise \(C1\)](#)). Replacing ρ by its tangent at a preliminary estimate gives a weighted lasso (Zou and Li, 2008), as does the adaptive lasso (Zou, 2006). Fan and Li (2001) proved an *oracle property*, which we state without proof: with p fixed, $\lambda_n/n \rightarrow 0$, $\lambda_n/\sqrt{n} \rightarrow \infty$ (in the scaling of [Eq. 30.1.1](#); Fan and Li write the penalty as $n \sum_j p_\lambda(|b_j|)$, where the conditions read $\lambda_n \rightarrow 0$, $\sqrt{n}\lambda_n \rightarrow \infty$) and regularity conditions, some local SCAD minimizer finds the zero coefficients with probability tending to one and estimates the rest as well as least squares on the true support. The convergence is not uniform in β (Leeb and Pötscher, 2005), and it does not justify oracle standard errors ([Chapter 29](#)).

30.1.6 Exercises

30.1.6.1 A. Check your understanding

Exercise A1.

With $\lambda = 1$, compute the lasso, ridge (penalty $\frac{1}{2}b^2$), hard, firm ($\gamma = 3$) and SCAD ($a = 3.7$) estimates of θ for a single observation $z = 2.5$ under an orthonormal design. Which rules shrink it, and by how much?

Solution. Lasso: $2.5 - 1 = 1.5$. Ridge: $2.5/2 = 1.25$. Hard: 2.5 , since $|z| > 1$. Firm: $\lambda < z \leq \gamma\lambda = 3$, so $(2.5 - 1)/(1 - 1/3) = 2.25$. SCAD: $2 < z \leq 3.7$, so $(2.7 \cdot 2.5 - 3.7)/1.7 = 3.05/1.7 = 1.79$. Only hard thresholding leaves z alone. At $z = 4$ the firm and SCAD rules would leave it alone as well, since 4 exceeds both $\gamma\lambda = 3$ and $a\lambda = 3.7$.

Exercise A2.

Show that the firm thresholding rule ϕ of [Proposition 30.1.5\(b\)](#) is continuous at $|z| = \lambda$ and at $|z| = \gamma\lambda$, whereas the hard thresholding rule $z\mathbf{1}(|z| > \lambda)$ jumps by λ at $|z| = \lambda$. What is the largest slope of ϕ , and what happens to it as $\gamma \downarrow 1$?

Solution. Take $z \geq 0$. As $z \downarrow \lambda$ the middle branch $(z - \lambda)/(1 - 1/\gamma)$ tends to 0, the value below λ . At $z = \gamma\lambda$ it equals $(\gamma - 1)\lambda/\{(\gamma - 1)/\gamma\} = \gamma\lambda$, the value of the upper branch. Hard thresholding is 0 at $z = \lambda$ and tends to λ as $z \downarrow \lambda$. The slope of ϕ is 0, then $\gamma/(\gamma - 1)$, then 1; the largest is $\gamma/(\gamma - 1)$, which tends to infinity as $\gamma \downarrow 1$, where ϕ approaches the discontinuous hard rule.

30.1.6.2 B. Practice

Exercise B1.

Derive the SCAD thresholding rule displayed above. Minimize $g(\theta) = \frac{1}{2}(z - \theta)^2 + \rho(\theta)$ for $z \geq 0$ on the three pieces $[0, \lambda]$, $[\lambda, a\lambda]$ and $[a\lambda, \infty)$, and check that g is convex on $[0, \infty)$ when $a > 2$.

Solution. On $(0, \infty)$, $g'(\theta) = \theta - z + \rho'(\theta)$. This is $\theta - z + \lambda$ on $(0, \lambda]$, $\theta - z + (a\lambda - \theta)/(a - 1)$ on $(\lambda, a\lambda]$, and $\theta - z$ beyond. The middle piece has slope $1 - 1/(a - 1) = (a - 2)/(a - 1) > 0$ when $a > 2$, and the pieces agree at λ and at $a\lambda$. So g' is continuous and increasing, and g is strictly convex on $[0, \infty)$. As in the proof of [Proposition 30.1.5](#), a negative θ is never better when $z \geq 0$. If $z \leq \lambda$, then $g'(0+) = \lambda - z \geq 0$ and $\hat{\theta} = 0$. If $\lambda < z \leq 2\lambda$, the root $z - \lambda$ lies in $(0, \lambda]$. If $2\lambda < z \leq a\lambda$, the middle piece vanishes at $\theta = \{(a - 1)z - a\lambda\}/(a - 2)$, which lies in $(\lambda, a\lambda]$ exactly for these z . If $z > a\lambda$, then $\hat{\theta} = z$.

Exercise B2.

Let P be any penalty on the slopes, and let the model have an unpenalized intercept. By [Proposition 27.2.1](#), $(\hat{b}_0, \hat{\beta})$ minimizes $\frac{1}{2} \sum_i (y_i - b_0 - \mathbf{x}_{(i)}^\top \mathbf{b})^2 + \lambda P(\mathbf{b})$ iff $\hat{\beta}$ minimizes

the centred criterion and $\hat{b}_0 = \bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}$. Deduce that the fitted values of penalized regression with an intercept are unchanged when a constant is added to $\mathbf{y}$, and that the slopes are unchanged when a constant is added to a column of $\mathbf{X}$. Does the second statement survive standardizing the columns before fitting?

30.1.6.3 C. Going deeper

Exercise C1.

Let $n = p = 2$, $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ with $r \in (0, 1)$, and use MCP with $\gamma > 1$. For which r does [Proposition 30.1.5\(a\)](#) guarantee convexity? Find a response $\mathbf{y}$ for which the MCP criterion has two distinct local minima when r exceeds that bound.